

NETWORKS AND COMPLEXITY

Exercise Sheet 22: Stripes on a Tiger

*This is an exercise sheet from the forthcoming book *Networks and Complexity*.
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

Ex 22.1: Kronecker products [Lvl. 1]

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 4 \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (1)$$

Compute a) $\mathbf{A} \otimes \mathbf{B}$, b) $\mathbf{A} \otimes \mathbf{C}$ and c) $\mathbf{C} \otimes \mathbf{A}$

Ex 22.2: Kronecker rules [Lvl. 1]

Show either generally or with the matrices from the previous exercise that

$$\text{a) } c(\mathbf{A} \otimes \mathbf{C}) = c\mathbf{A} \otimes \mathbf{C} = \mathbf{A} \otimes c\mathbf{C}, \text{ b) } (\mathbf{A} \otimes \mathbf{C})(\mathbf{u} \otimes \mathbf{w}) = \mathbf{A}\mathbf{u} \otimes \mathbf{C}\mathbf{w},$$

where c is a scalar and $\mathbf{u}$ and $\mathbf{w}$ are vectors of suitable dimension.

Ex 22.3: General two-dimensional system [Lvl. 2]

Consider a network, where the dynamics of the nodes is given by

$$\begin{aligned} \dot{x}_i &= f(x_i, y_i) - c_x \sum L_{ij} x_j \\ \dot{y}_i &= g(x_i, y_i) - c_y \sum L_{ij} y_j \end{aligned}$$

where x_i, y_i are the local variables in node i , f and g are arbitrary functions, $\mathbf{L}$ is the network Laplacian and c_x, c_y are diffusion rates.

- a) Show that, if (x^*, y^*) is the steady state of a network containing only a single node, then every network has a homogeneous steady state at $x_i = x^*, y_i = y^*$ for all i .
- b) Compute the Jacobian $\mathbf{J}$ for this system on the linked pair and show that it can be written in the form $\mathbf{J} = \mathbf{I} \otimes \mathbf{P} - \mathbf{L} \otimes \mathbf{C}$, where $\mathbf{P}$ is the Jacobian of the a network consisting of a single isolated node.

Ex 22.4: Abstract stability analysis [Lvl. 2]

We place copies of a dynamical system into the nodes of a network and couple them diffusively such that the local within-node Jacobian and the coupling matrix are given by

$$\mathbf{P} = \begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (41)$$

Compute the master stability function for this system. Describe what the result means for the network.

Ex 22.5: Product Formula [Lvl. 3]

Consider the Kronecker product

$$\mathbf{J} = \mathbf{A} \otimes \mathbf{B}. \quad (57)$$

Find a formula that relates the eigenvalues of $\mathbf{J}$ to the eigenvalues of $\mathbf{A}$ and $\mathbf{B}$.

Ex 22.6: A predator-prey model [Lvl. 3]

A predator-prey system with quadratic mortality can be written as

$$\begin{aligned}\dot{x} &= (4x^2 + x)/5 - xy \\ \dot{y} &= xy - y^2\end{aligned}$$

- Compute the positive steady state of the system and the corresponding Jacobian $\mathbf{P}$.
- Now consider a geographical system, consisting of several habitat patches. Each patch internally has the dynamics from a). The patches form a geographical network, where the links correspond to avenues of dispersal. The prey disperses diffusively between these patches at rate $1/5$. The predator also disperses diffusively but at rate 1 . Find the critical values of the Laplacian eigenvalue κ at which Turing bifurcations of the positive homogeneous state occur.
- Show that wave instabilities cannot occur.
- Consider a linked pair of nodes, where the link has weight w . For which link weights is the positive homogeneous state stable?

Ex 22.7: Flight response [Lvl. 3]

Consider again the predator-prey system from the previous exercise. But now the predator, y , does not diffuse, while the prey x tries to avoid the predators. Assume that the departure rate of prey individuals from a patch is axy , where a is a parameter.

- Consider the system on the linked-pair network. Write the differential equations and show that the steady state from the previous exercise is still stationary.
- Show that the Jacobian for the linked pair can still be written in the form $\mathbf{J} = \mathbf{I} \otimes \mathbf{P} - \mathbf{L} \otimes \mathbf{C}$, albeit with a different matrix $\mathbf{C}$.
- Is the homogeneous state stable in this case?

Ex 22.8: Two nice results [Lvl. 4]

Show that the following holds for any reaction diffusion system:

- A steady state that is unstable in a single isolated node can never be stabilized by coupling multiple identical nodes together.
- Diffusion can never destabilize a homogeneous state if all variables diffuse independently and at the same rate, i.e. $\mathbf{C} = c\mathbf{I}$.

Ex 22.9: Spectra of hypercubes [Lvl. 4]

A hypercube is a generalization of cubes to arbitrary dimensions. The 0-dimensional hypercube is just a single node.

- Compute the eigenvalue of the adjacency matrix of the 0-dimensional hypercube, $\mathbf{A}_0$.
- To make a 1-dimensional hypercube we take the 0-dimensional hypercube and make a copy of it. Then we connect every node in the original cube with the same node in the copy. Compute the eigenvalues of the corresponding adjacency matrix $\mathbf{A}_1$.
- To make a 2-dimensional hypercube we take the 1-dimensional hypercube and make a copy of it. Then we connect every node in the original 1D-cube with the same node in the copy. Try to express the adjacency matrix of the 2-dimensional hypercube, $\mathbf{A}_2$, as a function of $\mathbf{A}_1$, $\mathbf{I}$ and

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Note that $\mathbf{M}$ plays a different role than $\mathbf{A}_2$. If you use $\mathbf{M}$ in the right way you will have an equation that also relates $\mathbf{A}_0$ and $\mathbf{A}_1$ if you shift the index. Once you have convinced yourself of your formula, use it to compute the eigenvalues of $\mathbf{A}_2$.

- d) To make a 3-dimensional hypercube we take the 2-dimensional hypercube and make a copy of it. Then we connect every node in the original 2D-cube with the same node in the copy. Check if your formula for the procedure still works, then use it to compute the eigenvalues of $\mathbf{A}_3$.
- e) Compute the Laplacian eigenvalues of the 3-dimensional hypercube; call it $\mathbf{L}_3$
- f) Is there a general rule for the Laplacian eigenvalues of hypercubes? Can you use it to compute the Laplacian eigenvalues of the 4-dimensional hypercube?
- g) Now that we have spotted the pattern, consider the generating function

$$G_i(x) = \sum c_{in} x^n$$

where c_{in} is the multiplicity of the eigenvalue $\lambda = n$ in the Laplacian spectrum of the i -dimensional hypercube. Write an iteration rule that relates G_i and G_{i+1} and hence find a closed form for G_i .

- h) Use your result to compute the sum of all Laplacian eigenvalues of the 1000-dimensional hypercube. Is there a way in which we can test if the result is correct?

Ex 22.10: Working backwards [Lvl. 3]

Suppose we wanted to make a dynamical system that exhibits pattern formation if placed on a triangle but not on many other networks. Our starting point is the observation that the triangle has a Laplacian eigenvalue $\kappa = 3$.

- a) Define the node Jacobian and the coupling matrix as

$$\mathbf{P} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 0 \\ 0 & 10 \end{pmatrix}$$

Demand that the system has a Turing bifurcation at $\kappa = 2.5$ and another one at $\kappa = 3.5$. Use the resulting conditions to express d and bc as a function of a

- b) Ensure that your system is stable for $\kappa = 0$ but unstable for $\kappa = 3$. Choose your values for a and b accordingly, then compute c and d . [Avoid computing eigenvalues if you can.]
- c) You should now have a fully specified node Jacobian $\mathbf{P}$. Now design a dynamical system $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ for which $\mathbf{P}$ appears as the Jacobian in a steady state.

Ex 22.11: Jacobian derivation [Lvl. 4]

In the chapter we used a system-level approach to compute the Jacobian of a general system where the dynamics of variable n on node i is

$$\dot{x}_{ni} = f_n(x_{:i}) - c_n \sum L_{ij} x_{nj}$$

- a) Compute Jacobian elements using the node-level approach

$$\mathbf{J}_{ni,mj} = \left. \frac{\partial \dot{x}_{ni}}{\partial x_{mj}} \right|_*$$

and show that it can still be written as $\mathbf{J} = \mathbf{I} \otimes \mathbf{P} - \mathbf{L} \otimes \mathbf{C}$

- b) Compute the Jacobian again using the intermediate blockwise approach

$$\mathbf{J}_{ij} = \left. \frac{\partial \dot{\mathbf{x}}_{:i}}{\partial \mathbf{x}_{:j}} \right|_*$$

Ex 22.12: Pattern formation in continuous space [Lvl. 4]

Consider the partial differential equation (PDE) system

$$\begin{aligned}\dot{x}(l) &= f(x(l), y(l)) + c_x \Delta x(l) \\ \dot{y}(l) &= g(x(l), y(l)) + c_y \Delta y(l)\end{aligned}$$

where $x(l)$ and $y(l)$ are defined in a one-dimensional space parameterized by the spatial coordinate l , such that $0 \leq l \leq 1$ and $\Delta = \partial^2 / \partial l^2$.

- a) Show that if (x^*, y^*) is the steady state of the non-spatial system, $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$, then $x(l) = x^*$, $y(l) = y^*$ is a solution of the PDE.
- b) Consider a small perturbation $\boldsymbol{\delta} = (\delta_x(l), \delta_y(l))^T$ from the steady state. Linearize the functions f and g , and define $\mathbf{I}$, $\mathbf{P}$, $\mathbf{C}$, $\mathbf{x}$ and $\otimes$ in such a way that you can write the linearized system as $\dot{\boldsymbol{\delta}} = (\mathbf{I} \otimes \mathbf{P} + \Delta \otimes \mathbf{C})\boldsymbol{\delta}$.
- c) Identify the Jacobian and find an efficient way to compute its eigenvalues, using the boundary conditions $\boldsymbol{\delta}(0) = \boldsymbol{\delta}(1) = 0$.